

Characterizing Persuasion Patterns in Climate Discourse on Brazilian Portuguese YouTube Videos

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Abstract

While climate change remains a critical global challenge, research on online climate discourse has been constrained to English-language content, hindering a comprehensive understanding of its global dynamics. This study analyzes 227,159 Brazilian Portuguese YouTube videos to identify persuasive characteristics of climate communication and key discussion themes and players. We find that persuasion strategies rooted in emotional appeal, moral framing, and reciprocity are positively correlated with audience engagement, while logic-based arguments, statistical evidence, and authority narratives exhibit negative influences. We identified 25 key topics and 6 major channel themes, representing the heterogeneous nature of online climate discussion. We discuss the implications of our findings, highlight the potential risks of public opinion manipulation using AI, and outline future research directions.

CCS Concepts

• Applied computing → Psychology; • Human-centered computing → Collaborative and social computing.

Keywords

Climate Change, YouTube, Persuasion, Brazilian Portuguese

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1 Introduction

Climate change is one of the critical global challenges identified in the United Nations Sustainable Development Goals (SDG #13: Climate Action) and has profound implications for public health, food security, economic stability, and geopolitical dynamics [27]. In response, policymakers and international organizations have engaged in extensive negotiations, summits, and treaty discussions to devise effective mitigation strategies. A key forum for these efforts is the United Nations Framework Convention on Climate Change (UNFCCC), commonly known as the Conference of the Parties (COP), which provides a platform for assessing progress, proposing agreements, and shaping international climate governance [17].

At the grassroots level, climate discussions occur in everyday rhetoric and on social media platforms, influencing political will, policy adoption, and international cooperation [2]. One such example is the climate activist Greta Thunberg, who gained momentum through social media and ultimately led to her speech at COP24 [24]. In this vein, social media data offer a valuable lens through which to explore public attitudes, serving as proxies for understanding public perception on critical social issues such as climate change [6]. In particular, YouTube is a platform characterized by extensive climate-related discussions and a substantial user base [23]. It also serves as an information source, with media organizations, policymakers, and advocacy groups leveraging its reach to facilitate communication and foster public involvement.

Despite the growing relevance of digital platforms, research on online climate discourse has remained largely confined to English-language content [1, 23] and limited keywords [11, 14], restricting efforts to develop a comprehensive understanding of global climate conversations. Brazil, an important developing country in the Global South and host of the COP30, plays a central role in the dialogue about climate change, due to its stewardship of the Amazon rainforest and its vast freshwater reserves. Furthermore, the rise of AI-generated content [3] further complicates the landscape, especially since such content has been shown to shape public beliefs about climate change [10]. Understanding the persuasive strategies used in online climate discussions is important, as public opinions can be subject to manipulation.

Building upon existing research, we collected 227,159 Brazilian Portuguese YouTube videos and analyzed 10 persuasion strategies drawn from the psychological literature to study their effectiveness in driving user engagement. Our findings reveal that emotional appeal, moral appeal, and reciprocity are positively correlated with video engagement, whereas logical appeal, statistical evidence, and authority show negative correlations with user engagement patterns. To explore the dynamics of climate-related discourse, we identified 25 key topics and six influential channels, examining the interplay between persuasion strategy, discussion topic, content creator, and their influence on user engagement. We discuss the implications of our findings, outline future research directions, and offer insight into the evolving role of AI-generated content in climate governance and its impact on public participation in global climate summits.

2 Related Work

2.1 Online Climate Discourse

Examining online conversations on socially significant topics like climate change is crucial for both policymaking and understanding public sentiment. Shapiro and Park [22] analyzed YouTube comments on climate-related videos to explore politicization and derive political insights. Allgaier [1] investigated climate-related web searches to assess the spread of conspiracy theories. Furthermore, research has shown that climate discussions are often dominated by a small group of highly active users [23], underscoring the importance of evaluating the influence these users have on discourse and ensuring a balanced representation of perspectives in climate-related conversations. While these studies provide valuable perspectives on climate discourse, they predominantly focus on English-language content. Our research expands on these findings by exploring climate conversations in Brazil, a key developing nation in the Global South, contributing to a more holistic understanding of global climate communication.

The existing examination of climate rhetoric is also generally limited to specific topics, restricting the analysis to subsets of the general discussion. Jang and Hart [14] examined polarization in climate discussions on Twitter and found that right-leaning groups prefer the term “Global Warming” over “Climate Change.” Falkenberg et al. [11] examined Twitter discussions on COP keywords from 2014 to 2021, finding increasing polarization driven by right-wing opposition to climate policies after 2019. Pera and Aiello [18] identified eight climate topics on YouTube and TikTok and found that the comments on YouTube were more aligned with the video content. Our work also contributes to ongoing research by identifying key themes and actors in large-scale online discussions.

2.2 Climate Belief Persuasion

Persuasion research has deep roots in social psychology and marketing. Foundational theories such as Cialdini’s principles of influence [8] identify key mechanisms (e.g., reciprocity, social norm) that underlie effective persuasive appeals. In applied domains such as advertising, these mechanisms manifest themselves as concrete strategies. For example, customer behaviors are influenced through emotional appeals and credibility cues in advertisements [16]. Researchers have expanded these insights to science communication

Table 1: Sample Brazilian Portuguese YouTube search keywords (English translations in parentheses).

Theme	Example Keywords
Climate change	Mudança climática (<i>climate change</i>), aquecimento global (<i>global warming</i>).
Climate policy	COP30, acordo de Paris (<i>Paris agreement</i>).
Sustainability	Reciclagem (<i>recycling</i>), sustentável (<i>sustainable</i>).
Climate denial	Resfriamento global (<i>global cooling</i>), aquecimento global é falso (<i>global warming is fake</i>).

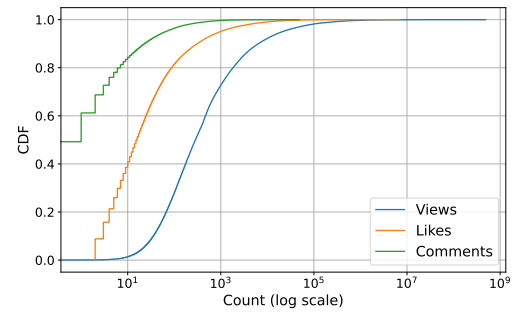


Figure 1: Cumulative distribution function for view, like and comment count for the collected videos.

and conspiracy contexts, investigating how framing and targeted messaging can shape public beliefs.

Within climate communication, persuasion is increasingly recognized as a powerful tool for driving positive social impact. Czarnek et al. [10] demonstrate how personalized, evidence-based dialogues can effectively reduce climate change skepticism. Likewise, AI-driven persuasive interventions, such as chatbot interactions, have been shown to diminish users’ belief in conspiracies [9]. As automated systems can now generate persuasive content [13], understanding the effectiveness of these strategies is more crucial than ever. Building on previous research, this study explores the dynamic relationship between persuasive climate messaging and user engagement patterns.

3 Data Methodology

We gathered climate-related videos on YouTube using a set of 65 keywords commonly found in Brazilian social media. These keywords were identified based on prior research [4, 20] and established climate change glossaries [19, 26]. They encompass a wide range of themes, including general climate change discussions, policy-related terminology, sustainability concepts, and climate change denial phrases, among others, as examples shown in Table 1.

We used YouTube Data API v3 to collect video metadata from the search list endpoint based on highest relevance. We set “preferred language” as Brazilian Portuguese and “preferred location” as Brazil to maximize the context relevance when querying the curated list of keywords. To ensure the analysis encompasses a broad timeframe, we collected videos spanning from January 1,

Table 2: Statistics of the collected videos. Note that the mean is skewed due to some videos being extremely popular.

	Mean	Median	Total
Views	17,058	147	3.87×10^9
Likes	496	8	1.13×10^8
Comments	18	0	4.07×10^6
Duration (min)	20	4	4.56×10^6

2019, to May 1, 2025. For each video, we extracted key information, including publication time, video ID, title, description, view count, like count, comment count, transcription (if available), channel ID, channel title, duration, and tags. On average, each keyword yielded approximately 24,000 videos. After removing duplicates, the initial dataset comprised 1,236,148 unique videos.

For geographical relevance, we first applied fastText language identification [15] to filter the collected data and identified 458,276 videos in Portuguese-language. Next, we also assessed their topic relevance to climate discussions using a large language model (GPT-4.1-mini-2025-04-14) with a temperature of 0 to minimize variability. This classification yielded a subset of 244,593 videos explicitly related to climate change. The distinction between keyword-based retrieval and language model relevance confirmation is essential, as certain keywords carry multiple meanings (e.g., "climate" in Portuguese can refer to a psychological atmosphere). Finally, we excluded videos with missing values in view, comment, and like counts, resulting in a final dataset of 227,159 videos.

Table 2 presents a summary of the data. There is a strong contrast between the mean and median statistics for the videos, suggesting that the attention of users is not well distributed on the platform, being concentrated on a small percentage of videos: 1.8% of the videos account for 90% of the views, while 6.8% of videos account for 90% of the comments. Figure 1 further shows the skewed distribution of views, likes, and comments using the cumulative distribution function (CDF). Nearly 50% of the videos in our dataset receive no comments, which may indicate a lack of popularity. However, this could also be due to comments being disabled or the videos belonging to YouTube Kids, where commenting is restricted. Video views show a strong correlation with likes (Spearman $\rho=0.89$) and a high correlation with comments ($\rho=0.61$), underscoring the relationship between user engagement metrics.

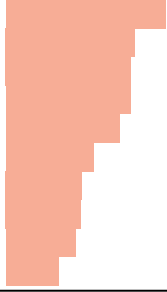
4 Results

We first examine the linguistic characteristics of persuasion strategies and subsequently analyzes their differential application across (1) discussion topics and (2) content creators. To quantify user engagement, two indices—like ratio and comment ratio—are introduced. These metrics are computed by normalizing the number of likes and comments relative to video views, respectively, providing a measure of the probability that a viewer will engage with a video either by liking or commenting post-exposure.

4.1 Persuasion Narrative

Identifying effective persuasion strategies in YouTube video engagement patterns suggests possible risks of opinion manipulations. To

Table 3: Persuasion strategy distribution shown by the number of videos employing each strategy and their corresponding percentage of the overall dataset.

Persuasion Strategy			
Logical Appeal	109,395		48.2 %
Common Ground	86,610		38.1 %
Authority	83,373		36.7 %
Emotional Appeal	82,981		36.5 %
Moral Appeal	74,529		32.8 %
Statistical Evidence	54,638		24.1 %
Social Norm	46,074		20.3 %
Personal Stories	45,244		19.9 %
Scarcity	41,095		18.1 %
Reciprocity	28,111		12.4 %

this end, we first curated 10 persuasion strategies in the context of climate change discourse based on previous literature [8, 9, 16]:

- **Logical Appeal:** Uses reasoning, evidence, and cause-and-effect arguments to explain climate change processes. Example description from the studied YouTube videos: *"Wildfires in Brazil have devastated our forests, putting biodiversity at risk and contributing to the climate crisis."*
- **Emotional Appeal:** Elicits emotions such as fear, guilt, or empathy to draw feelings about climate events. Example: *"The fear is that if the scenario doesn't change, the world as we know it today could end."*
- **Statistical Evidence:** Relies on concrete data, research, and environmental statistics. Example: *"2023 is already the hottest year in the last 125,000 years."*
- **Social Norm:** Refers to what others are doing or what is socially accepted to create peer pressure. Example: *"Two hundred countries have signed an agreement to control the production..."*
- **Authority:** Using experts, institutions, or official reports to gain trust. Example: *"Marina Silva emphasized the themes of environmental justice and the GDP from ecosystem services"*
- **Personal Stories:** Uses individual experiences to illustrate climate impacts. Example: *"I became a garbage collector for a day with a group of women who handle our recyclable waste!"*
- **Moral Appeal:** Appeals to ethical duty, justice, or responsibility for nature. Example: *"Humanity is suffering from increasingly extreme droughts, floods, and heatwaves."*
- **Reciprocity:** Highlights mutual benefits and giving back to nature as a way to protect the resources. Example: *"Without the Amazon, there is no agriculture in the South and Southeast."*
- **Scarcity:** Presents limited time and irreversible consequences to demand immediate climate action. Example: *"Planet Earth has only 18 MONTHS to be SAVED!!"*
- **Common Ground:** Builds on shared identity, values, or national pride to create a sustainable future for all. Example: *"Let's build a greener, more prosperous future together!"*

We used gpt-4o-mini-0718 to identify the persuasion strategy used in YouTube videos. Validation was done independently, using the combination of video title, description, and transcription (if it

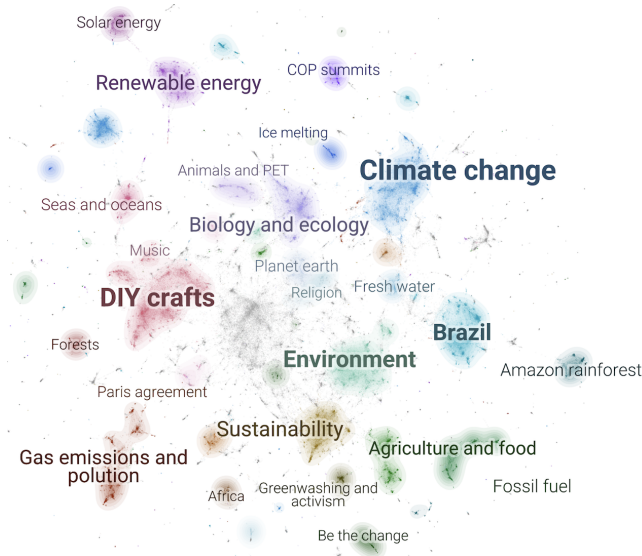


Figure 2: Visualization of climate topic clusters, showing video content distributed across general themes and context-specific topics like the Amazon rainforest and Brazil.

exists) as input. The above definitions were included as a system prompt, with 5-shot examples used for each definition to enhance context. The temperature was set to 0 to minimize the variability in the output. The predictions achieved 90.6% accuracy against the consensus annotations of two authors on 100 random samples. For robustness check, we examined videos with and without video transcriptions and found consistent distributions. Table 3 shows the persuasion strategies used. The most commonly used strategy is logical appeal, employed in 48.2% of videos to convey climate-related information based on logical reasoning.

To assess the impact of persuasion strategies on user engagement, we compared videos that employed specific strategies with those that did not. We performed t-tests to examine the effect of each persuasion strategy on two proposed engagement metrics: video like ratio and video comment ratio. Three strategies, logical appeal, statistical evidence, and authority, consistently led to lower user engagement ($ps < 0.05$). For instance, the use of authority and credibility in video content resulted in a like ratio of 8.00% and a comment ratio of 1.38%, whereas videos that did not employ authority to convey climate-related information received a like ratio of 8.97% and a comment ratio of 2.37%. In contrast, three strategies (moral appeal, emotional appeal, and reciprocity) consistently facilitated engagement through higher likes and comments ($ps < 0.05$). Taking moral appeal as an example, this strategy was associated with significantly higher levels of like ratio (9.61% in videos using moral appeal vs. 8.13% in those that did not) and comment ratio (2.26% vs. 1.89%, respectively).

4.2 Thematic Analysis

We used BERTopic [12] to extract interpretable topics from YouTube video titles. This algorithm is well suited for social media posts to encode semantic meaning from short text. These embeddings

Table 4: Top and bottom 5 topics ranked by like and comment ratios.

Like Ratio	Ratio	Comment Ratio	Ratio
Top 5			
Be the change	19.71%	Be the change	8.16%
Religion	11.07%	Planet earth	2.46%
Planet earth	9.49%	Religion	2.22%
Biology and ecology	9.48%	Climate change	2.08%
Climate change	9.38%	Music	2.01%
Bottom 5			
COP summits	7.11%	COP summits	1.21%
Fresh water	7.04%	Solar energy	1.11%
Fossil fuel	7.02%	Ice melting	1.08%
Solar energy	6.22%	Fresh water	0.98%
Africa	5.56%	Africa	0.80%

undergo dimensionality reduction via UMAP and are subsequently clustered using HDBSCAN, yielding semantically coherent topic representations.

The 25 extracted topics were labeled by native speakers based on video samples and representative keywords, as presented in Figure 2. We found that YouTube content spans a diverse array of themes beyond general climate-related issues, including social movements (e.g., “Be the Change”), activism and resistance to greenwashing, entertainment such as DIY content and music (e.g., climate protest songs), politics with explicit references to COP and the Paris Agreement, and religion (e.g., prayers for climate or Pope Francis’ speech on climate change).

From the persuasion aspect, different topics received varied user engagements. Table 4 shows five topics that were liked the most and least and commented on in relation to their views, where the social movement “Be the change” that attracted significantly more engagements. We contrast the persuasion strategies from that topic with the COP summits, one of the least engaging topics, to draw policy-relevant insights. Figure 3 shows the persuasion distributions

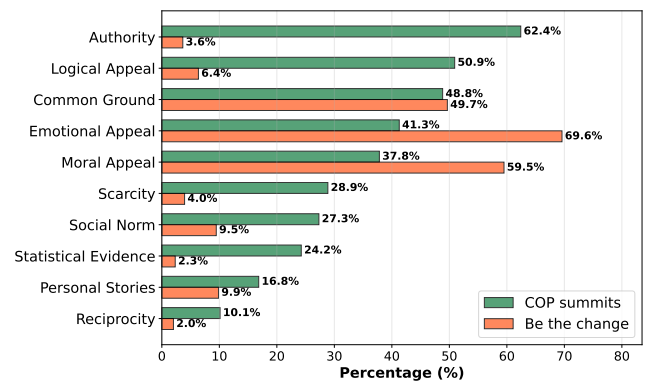


Figure 3: Distribution of persuasion strategies for the topics “COP summits” and “Be the change”. The primary persuasion strategies among COP summits are authority and logical appeal, while “Be the change” employs more emotional and moral appeals.

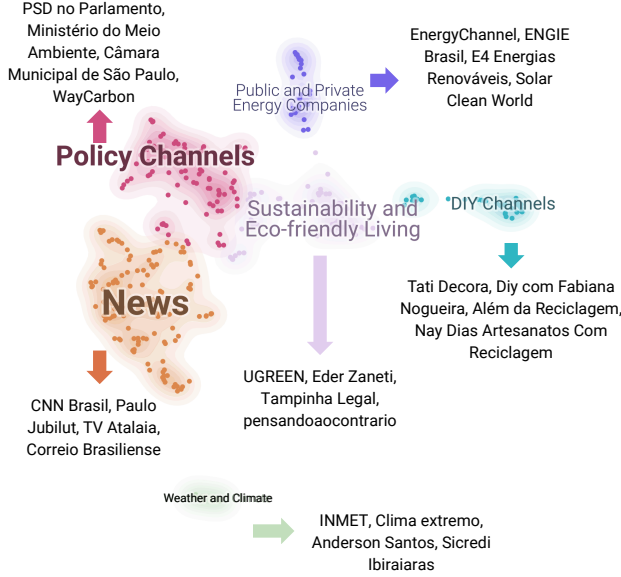


Figure 4: YouTube climate content is primarily clustered around news and policy channels, with individual creators forming smaller groups.

for the two topics, where “Be the change” correlates with more emotional and moral appeal strategies, while the COP topic tends to include more authority and logical appeal narratives, possibly due to its policy-driven context.

4.3 Content Creator

To analyze content from the most influential channels related to climate change keywords, we apply the tube2vec-con [5] method to generate representations for each eligible channel, consisting of the average embedding of the latest 50 video titles and descriptions. We reduced the dimensionality of embeddings using UMAP, followed by the application of the KMeans algorithm, labeling each cluster upon manual inspection. The number of clusters was based on the elbow method using the silhouette score.

Figure 4 presents the resulting channel clusters. We identify six distinct clusters: news, policy channels, energy companies, sustainability and eco-friendly living, DIY and weather and climate reporting. The majority of channels fall into the news (e.g., “CNN Brasil”) and policy categories (e.g., “City Council of São Paulo”), demonstrating their pivotal role in climate-related discourse. Within individual content creators, there are not only DIY craft makers, but also people reporting news, and educators, such as biologists and geographers, suggesting strong engagement with climate-related education on the platform. Given that education is a key factor in the response to climate change [25], the presence of such content is a valuable aspect of climate communication. These observations illustrate the heterogeneity of climate communication as people from vastly different backgrounds contribute to the discussion.

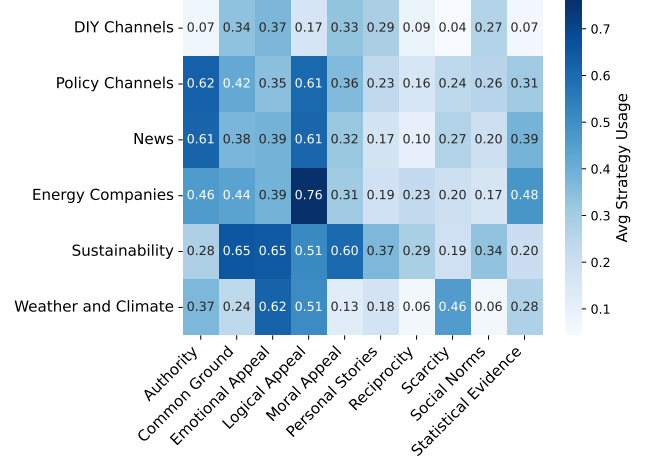


Figure 5: Average use of ten persuasion strategies across six channels. Energy companies use more logical appeals, while sustainability-focused channels employ more emotional, moral, and common ground rhetoric.

We show the usage of ten persuasion strategies across the six channel clusters in Figure 5. “Sustainability and Eco-friendly Living” on average uses more emotional and moral appeal methods in their YouTube content. It also has the highest like ratio as 12.17% and second highest comment ratio as 3.82%, corroborating our findings that moral and emotional strategies correlate to higher engagement.

5 Discussion and Conclusion

Climate Discourse Engagement. This study improves our understanding of climate public debate in Brazil, a major developing country in the Global South. Our analysis identifies key persuasion strategies and reveals a distinct pattern in audience engagement on YouTube. Specifically, fact-based persuasion methods, such as logic, statistics, and appeals to authority, tend to result in lower user engagement, measured by video likes and comments. In contrast, moral and emotional strategies, including appeals to emotion, morality, and reciprocity, are associated with significantly higher levels of engagement.

This trend remains consistent across different discussion topics and content creator channels. Our findings reaffirm that moral and emotional expressions are powerful in capturing attention and driving online interactions [7]. However, interpreting socially meaningful rhetoric presents new challenges in the era of generative AI [3]. Given that climate change is a pressing global issue and that digital content can influence beliefs [10], our research reinforces the link between persuasive content strategies and online behavioral patterns.

Limitations. Our study has several limitations. First, while we set preferred language and location parameters during YouTube video collection, our curated dataset may still include outlier samples that do not fully align with the intended context. Second, our findings represent correlational observations rather than definitive causal relationships between persuasion strategies and engagement.

metrics. We refrain from making strong claims about direct influences. Third, this study measures user engagement through the lens of video persuasions, which may not fully reflect the underlying psychological motivations.

Broader Social Impacts. This study is a preliminary exploration into public communication and participation patterns on climate change within the Global South. We hope that the findings are informative for policymakers, communication professionals, and NGOs. Historically, persuasion strategies have played a critical role in social movements such as tobacco, oil, food, and information networks. Our findings suggest that emotional appeals tend to drive engagement, implying the importance of decentralized individual influence in message dissemination on online media. This study suggests the potential for automated manipulation, particularly through recommendation algorithms that amplify persuasion strategies associated with higher engagement and influence. Given the persuasive nature of AI-generated content, society faces increasingly more challenges to ensure information veracity and risks of manipulation of public opinion [21].

Future Research Directions. We outline important future research directions that can further contribute to the understanding of public manipulation of socially sensitive discourse.

- **Dataset:** Future work may examine the content of user comments, where this work only uses the number of comments as a proxy of engagements. A thoroughly validated dataset of both video content and comments would enable understanding of relationships between video persuasion effects and user engagement patterns.
- **Misinformation:** We did not fact-check the video content, and user engagement may vary depending on whether the information presented is factual, false, or misleading. Examining misinformation dissemination through the lens of persuasion could offer new insights into safeguarding information integrity on socially meaningful discourses.
- **Scope:** Future work can expand our analysis to other languages and geographic areas to perform comparisons (e.g., English vs. Portuguese; Global North vs. Global South) and to focus on discussions about specific regions (e.g., oil exploration along the equatorial margin of the Amazon River).
- **Granularity:** Future work may investigate the influence of different combinations of persuasion strategies. We analyzed all videos over six years collectively. Measuring how persuasion strategies impact online behaviors overtime and how these changes reflect social events warrants further investigation.

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References

- [1] Joachim Allgaier. 2019. Science and environmental communication on YouTube: Strategically distorted communications in online videos on climate change and climate engineering. *Frontiers in Communication* 4 (2019), 36.
- [2] Ashley A. Anderson. 2017. Effects of Social Media Use on Climate Change Opinion, Knowledge, and Behavior. doi:10.1093/acrefore/9780190228620.013.369

- [3] Isabelle Augenstein, Timothy Baldwin, Meeyoung Cha, Tanmoy Chakraborty, Giovanni Luca Ciampaglia, David Corney, Renee DiResta, Emilio Ferrara, Scott Hale, Alon Halevy, et al. 2024. Factuality challenges in the era of large language models and opportunities for fact-checking. *Nature Machine Intelligence* 6, 8 (2024), 852–863.
- [4] Clara Baltasar, Sergio D'Antonio Maceiras, Alejandro Martín, and David Camacho. 2024. Analysis of Climate Change Misleading Information in TikTok. In *CEUR Workshop Proceedings*, Vol. 3782.
- [5] Léopaul Boesinger, Manoel Horta Ribeiro, Veniamin Veselovsky, and Robert West. 2024. Tube2Vec: Social and Semantic Embeddings of YouTube Channels. In *proc. of the ICWSM*.
- [6] Alexandre Bovet, Flaviano Morone, and Hernán A Makse. 2018. Validation of Twitter opinion trends with national polling aggregates: Hillary Clinton vs Donald Trump. *Scientific Reports* 8, 1 (2018), 8673.
- [7] William J Brady, Julian A Wills, John T Jost, Joshua A Tucker, and Jay J Van Bavel. 2017. Emotion shapes the diffusion of moralized content in social networks. *PNAS* 114, 28 (2017), 7313–7318.
- [8] Robert B Cialdini. 2001. The science of persuasion. *Scientific American* 284, 2 (2001), 76–81.
- [9] Thomas H Costello, Gordon Pennycook, and David G Rand. 2024. Durably reducing conspiracy beliefs through dialogues with AI. *Science* 385, 6714 (2024), eadq1814.
- [10] Gabriela Czarnek, Reed Orchinik, Hause Lin, Henry G Xu, Thomas Costello, Gordon Pennycook, and David G Rand. 2025. Addressing climate change skepticism and inaction using human-AI dialogues. doi:10.31234/osf.io/mqcwj_v1
- [11] Max Falkenberg, Alessandro Galeazzi, Maddalena Torricelli, Niccolò Di Marco, Francesca Larosa, Madalina Sas, Amin Mekacher, Warren Pearce, Fabiana Zollo, Walter Quattrociocchi, et al. 2022. Growing polarization around climate change on social media. *Nature Climate Change* 12, 12 (2022), 1114–1121.
- [12] Maarten Grootendorst. 2022. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794* (2022).
- [13] Kobi Hackenberg and Helen Margetts. 2024. Evaluating the persuasive influence of political microtargeting with large language models. *PNAS* 121, 24 (2024), e2403116121.
- [14] S Mo Jang and P Sol Hart. 2015. Polarized frames on “climate change” and “global warming” across countries and states: Evidence from Twitter big data. *Global environmental change* 32 (2015), 11–17.
- [15] Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. Bag of Tricks for Efficient Text Classification. In *proc. of the EACL*.
- [16] Yaman Kumar, Rajat Jha, Arunim Gupta, Milan Aggarwal, Aditya Garg, Tushar Malyan, Ayush Bhardwaj, Rajiv Ratn Shah, Balaji Krishnamurthy, and Changyou Chen. 2023. Persuasion strategies in advertisements. In *proc. of the AAAI*.
- [17] Jonathan Kuypers, Heike Schroeder, and Björn-Ola Linnér. 2018. The Evolution of the UNFCCC. *Annual Review of Environment and Resources* 43, 1 (2018), 343–368.
- [18] Arianna Pera and Luca Maria Aiello. 2024. Shifting climates: climate change communication from YouTube to TikTok. In *proc. of the ACM WebSci*. 376–381.
- [19] RSBP. 2024. Climate jargon buster - your guide to the key terms. <https://www.rspb.org.uk/helping-nature/what-we-do/influence-government-and-business/nature-and-climate-emergency/climate-change-key-words-explained>. Accessed: 2025-04-20.
- [20] Frederico Salmi and Lorena Cândido Fleury. 2022. Mudanças climáticas e Ciências Sociais: análise bibliométrica do campo (2011-2021). *Bib: revista brasileira de informação bibliográfica em ciências sociais*. 1, 97 (2022), 1–19.
- [21] Daniel Thilo Schroeder, Meeyoung Cha, Andrea Baronchelli, Nick Bostrom, Nicholas A Christakis, David Garcia, Amit Goldenberg, Yara Kyrychenko, Kevin Leyton-Brown, Nina Lutz, et al. 2025. How Malicious AI Swarms Can Threaten Democracy. *arXiv preprint arXiv:2506.06299* (2025).
- [22] Matthew A Shapiro and Han Woo Park. 2015. More than entertainment: YouTube and public responses to the science of global warming and climate change. *Social Science Information* 54, 1 (2015), 115–145.
- [23] Matthew A Shapiro and Han Woo Park. 2018. Climate change and YouTube: Deliberation potential in post-video discussions. *Environmental Communication* 12, 1 (2018), 115–131.
- [24] Amelia Tait. 2019. Greta Thunberg: How one teenager became the voice of the planet. <https://tinyurl.com/h9dmhaxm>. Accessed: 2025-05-27.
- [25] UNESCO. n.d.. Climate Change Education. <https://www.unesco.org/en/climate-change/education> Accessed: 2025-06-06.
- [26] UNFCCC. 2006. Glossary. https://unfccc.int/resource/cd_roms/na1/ghg_inventories/english/8_glossary/Glossary.htm. Accessed: 2025-04-15.
- [27] United Nations. [n. d.]. Causes and Effects of Climate Change. <https://tinyurl.com/2258h8yu>. Accessed: 2025-05-27.